

1 Summary

Model (Run-name): model1 (test)

Iteration: 31

TAC: 154888.44082960865

Took: 3.1226155757904053s

Total Time: 62.744773626327515s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

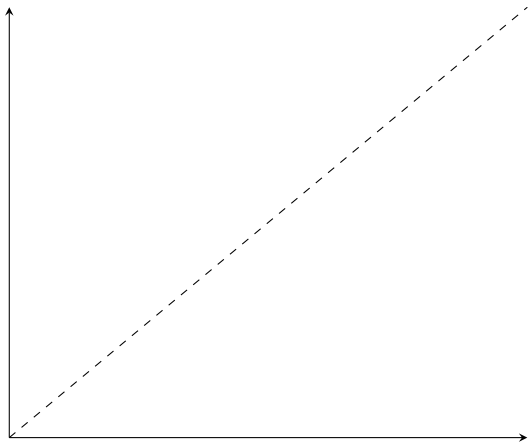
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

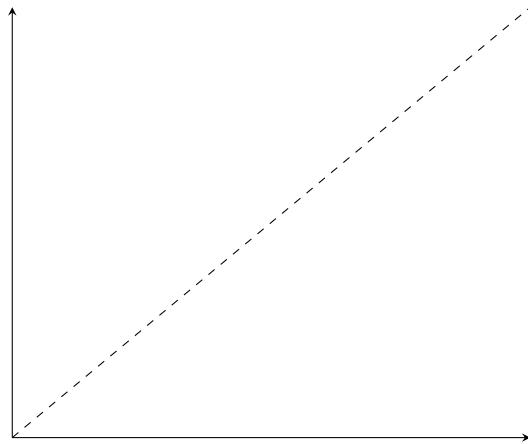
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052184765672136896	$b_{1,1,1}^{H,out}$	900.2515692555542
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017726451999674955	$b_{1,2,2}^{H,out}$	3868.728542184986
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028991606195275743	$b_{2,1,2}^{H,out}$	4836.832354611851
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014614638865694654	$b_{1,1,1}^{C,in}$	5238.350632985475
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009576840419405918	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022541417736144057	$b_{2,1,2}^{C,in}$	4151.0964249347835
$A_{1,1,1}^\beta$	71.03465240716494	$q_{1,1,1}$	681.271457815014	$b_{1,1,1}^{C,out}$	5919.622090800489
$A_{1,2,2}^\beta$	69.13316279873226	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	140.73764915547463	$q_{2,1,2}$	2428.0928132774793	$b_{2,1,2}^{C,out}$	6579.189238212262
$A_{cu,1}^\beta$	4.9192700692457025	$q_{cu,1}$	168.72854218498605		
$A_{cu,2}^\beta$	37.7631578546152	$q_{cu,2}$	1971.9071867225202		
$A_{hu,1}^\beta$	13.265751604070672	$q_{hu,1}$	490.635728907507		
$A_{1,1,1}$	71.03465240716494	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.13316279873226	$t_{1,2}^{(H)}$	581.8728542184987		
$A_{2,1,2}$	140.73764915547463	$t_{1,3}^{(H)}$	386.8728542184986		
$A_{cu,1}$	4.9192700692457025	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.7631578546152	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.265751604070672	$t_{2,3}^{(H)}$	468.595359336126		
$dt_{1,1,1}$	32.7090485938338	$t_{1,1}^{(C)}$	617.2909514061662		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.8728542184987		
$dt_{1,2,2}$	81.87285421849867	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.8728542184986	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.127145781501326	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.59535933612602	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.8728542184986	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.59535933612602	$t_{1,2,2}^H$	386.8728542184986		
$dt_{hu,1}$	62.7090485938338	$t_{2,1,2}^H$	384.04311670355423		
$f_{1,1,1}^H$	2.4331123493393356	$t_{1,1,1}^C$	644.128525725349		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	12.313432487948017	$t_{2,1,2}^C$	649.8205080141946		
$f_{1,1,1}^C$	9.160728897096938	$b_{1,1,1}^{H,in}$	1581.523027070568		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5818.7285421849865		
$f_{2,1,2}^C$	10.124625426670203	$b_{2,1,2}^{H,in}$	7264.92516788933		

3 *RecLMTD* Tangent Points

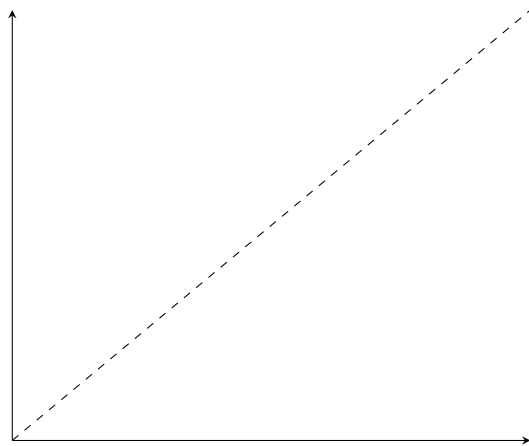
- $(1, 1, 1)$
 - Value: 0.052185
 - Error: 0.000000012441379
 - Relative error: 0.000000238410124
- $(1, 2, 2)$
 - Value: 0.017726
 - Error: 0.000000033851694
 - Relative error: 0.000001909667482
- $(2, 1, 2)$
 - Value: 0.028992
 - Error: 0.000000158365045
 - Relative error: 0.000005462414857
- $cu, 1$
 - Value: 0.014615
 - Error: 0.000000005603609
 - Relative error: 0.000000383424270
- $cu, 2$
 - Value: 0.009577
 - Error: 0.000000463722630
 - Relative error: 0.000048418910262
- $hu, 1$
 - Value: 0.022541
 - Error: 0.000000001319204
 - Relative error: 0.000000058523543



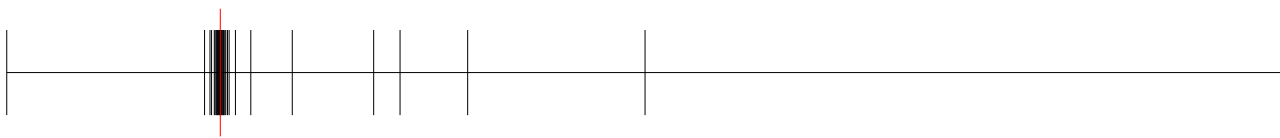
(a) *RecLMTD* - (1, 1, 1)



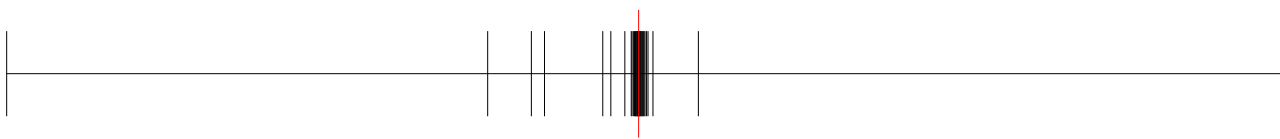
(b) *RecLMTD* - (1, 2, 2)



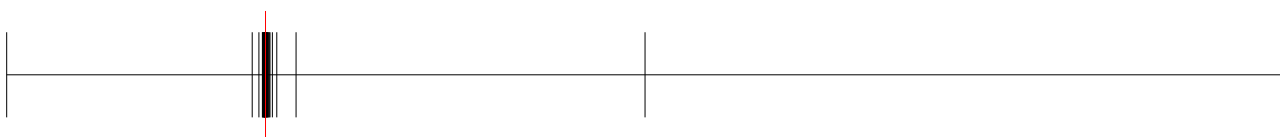
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1581.523027070568
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5818.7285421849865
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 7264.92516788933
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 900.2515692555542
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3868.728542184986
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 4836.832354611851
 - Absolute error: 107.94336462149477
 - Relative error: 0.02282636891032515

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5238.350632985475
 - Absolute error: 0.4215481192295556
 - Relative error: -8.046696910203553e-05
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4151.0964249347835
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 5919.622090800489
 - Absolute error: 18.935291743835478
 - Relative error: 0.003208997933403731
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6579.189238212262
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.3823811245472878e-16



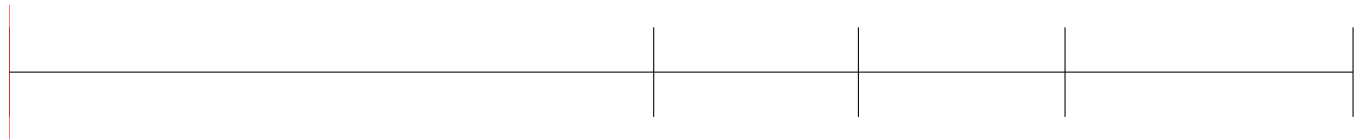
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



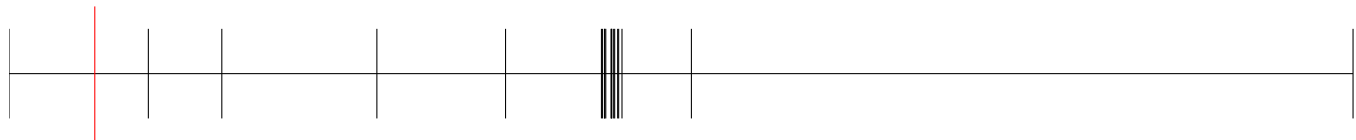
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$

5 Area Breakpoints

5.1 A

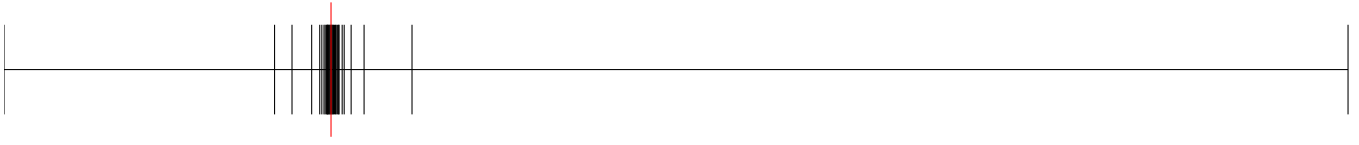
- $(1, 1, 1)$
 - Value: 71.034652
 - Error: 0.069330363218256
 - Relative error: 0.000975055974602
- $(1, 2, 2)$
 - Value: 69.133163
 - Error: 0.0000000000000071
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 140.737649
 - Error: 0.050972140765140
 - Relative error: 0.000362047303936

5.2 A_{CU}

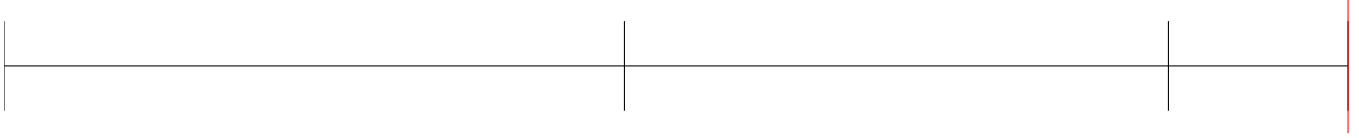
- $cu, 1$
 - Value: 4.919270
 - Error: 0.012543351491692
 - Relative error: 0.002543354831501
- $cu, 2$
 - Value: 37.763158
 - Error: 0.006123043627284
 - Relative error: 0.000162117029545

5.3 A_{HU}

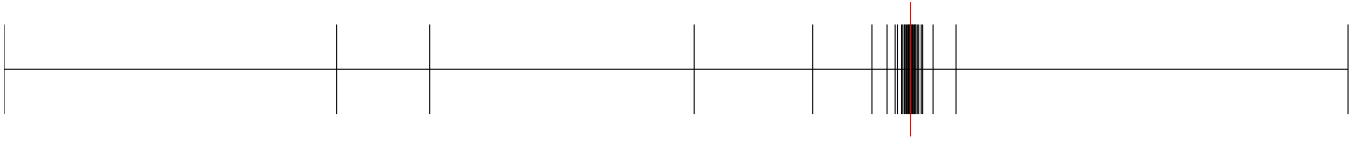
- $hu, 1$
 - Value: 13.265752
 - Error: 0.005798301827300
 - Relative error: 0.000436897112124



(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

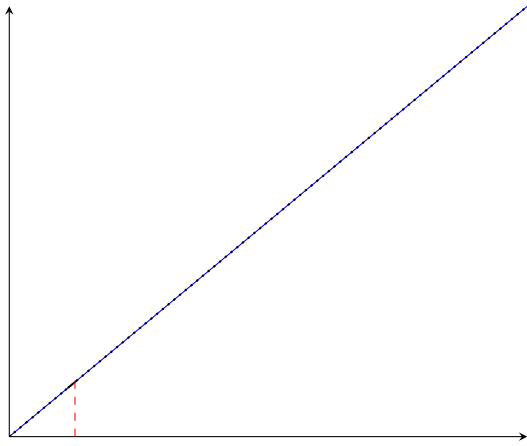


(a) $A^\beta - hu, 1$

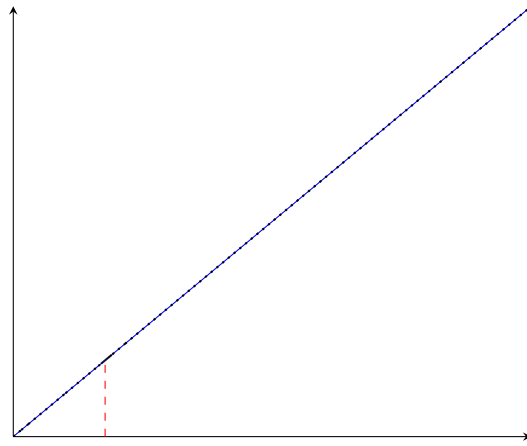
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.034652
 - $\hat{q}^\beta$: 71.034652
 - q^β : 71.034652
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.49758876855047, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.133163
 - $\hat{q}^\beta$: 69.133163
 - q^β : 69.133163
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 66.74637189255778, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 140.737649
 - $\hat{q}^\beta$: 140.737649
 - q^β : 140.737649
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 150.84969099235698, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.919270
 - $\hat{q}^\beta$: 4.919270
 - q^β : 4.919270

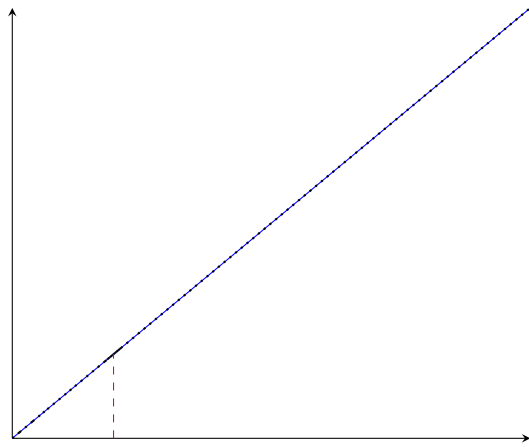
- Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, **4.888121952381175**, **4.962276317638335**, 4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.0892956597199035, 5.195781532076625, 5.3386847474643435, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.763158
 - $\hat{q}^\beta$: 37.763158
 - q^β : 37.763158
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.40128299925594, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65435035996605, 37.6551299822246, 37.682669717644174, **37.71812813120871**, **37.76995088018704**, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.265752
 - $\hat{q}^\beta$: 13.265752
 - q^β : 13.265752
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.064750414488246, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, **13.244985069124171**, **13.266498143734903**, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



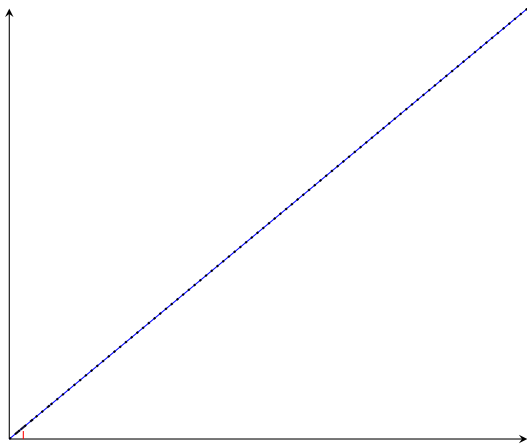
(a) $q^\beta - (1, 1, 1)$



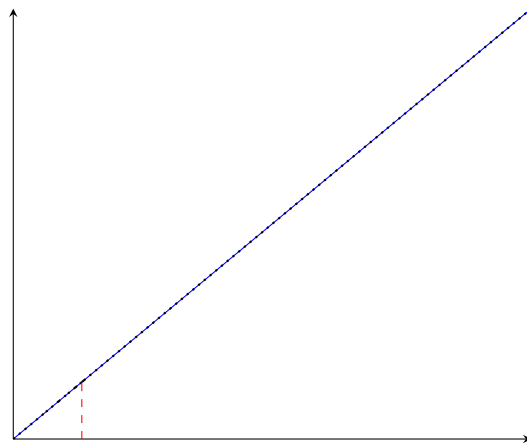
(b) $q^\beta - (1, 2, 2)$



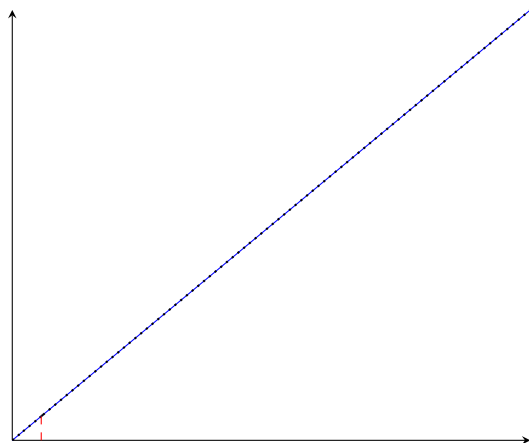
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	4535.074832110669
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1998.9035750652167
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008077568070374637	$b_{2,1,1}^{C,in}$	3339.7421802920035
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	1998.9035750652165
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	3339.7421802920035
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	563.3974443535717		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	587.8717680522652		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	16.039520885165338	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	562.3752572884988		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.0112035847521		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4918.476972929432		
$f_{1,2,1}^H$	7.566887650660664	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	4535.074832110669		
$f_{2,2,2}^H$	7.686567512051983	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.875374573329797	$b_{1,2,1}^{H,out}$	4918.476972929432		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	5.839271102903062	$b_{2,2,1}^{H,out}$	0.0		